Lab Eleven – Point Estimation and Resampling

Tips

- Complete the tasks below. Make sure to start your solutions in on a new line that starts with "**Solution**:".
- Make sure to use the Quarto Cheatsheet. This will make completing and writing up the lab *much* easier.
- Make sure to use "enter" to make your code readable, so it doesn't run off the page. I switched the Quarto document to automatically render to .pdf, so you can see what you're submitting by default. You can switch back for the highlighting by moving the html block above the pdf block in the YAML header.
- Don't forget that you can copy-and-paste code when you're doing something similar as we did in class!
- Make sure to plot all computed probabilities. This is good `ggplot` practice AND will help make sure you don't make a mistake when computing the probability.

1 Introduction

When performing point estimation, there isn't just one *right* answer. For example, the Method of Moments (MOM) and Maximum Likelihood Estimates (MLEs) aren't always the same.

One question that comes up, is why we calculate the sample variance as

$$S_{n-1}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2,$$

when it is supposed to be the "average squared distance from the mean, i.e.,

$$S_n^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

2 Comparing Estimators

2.1 Initial Simulation

In this lab, we will explore why we use S_{n-1}^2 and not S_n^2. To do this, we will generate data from four known distributions:

- Poisson($\lambda = 2$)
- Binomial$\left(n = 50, p = \frac{5-\sqrt{21}}{10}\right)$
- Normal($\mu = 0, \sigma = \sqrt{2}$)
- Exponential$\left(\lambda = \frac{1}{\sqrt{2}}\right)$.

Remark: Note that I have selected the parameters so that the true variance for all four models is 2.

For each distribution, write a loop that completes the following 1000 times. Put `set.seed(7272)` at the top of your code chunk.

- Generate a sample of size n from the distribution.
- Calculate and save S_{n-1}^2 and S_n^2.

Start with $n = 20$. Is there evidence that S_{n-1}^2 is a better estimator than S_n^2?

Create a plot or combination of plots (using patchwork) to show the results across all four distributions. Make sure to create a corresponding table that numerically summarizes the results of the simulation. You should include

$$\text{Bias} = \text{Average of Estimates} - \text{True Value}$$
$$\text{Precision} = \frac{1}{\text{Variance of Estimates}}$$
$$\text{Mean Square Error} = \text{Bias}^2 + \text{Variance of Estimates.}$$

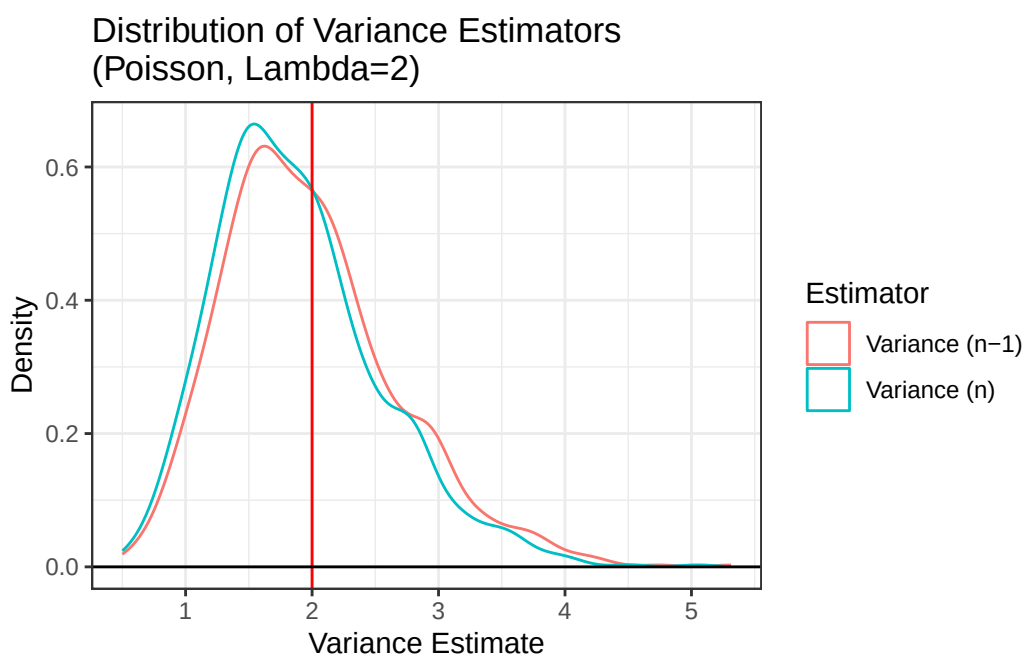
Solution:
Poisson:

```r
set.seed(7272)

var.n1 = NA #initialized vector for S_{n-1}^2
var.n = NA #initialized vector for S_{n}^2
n = 20
var.true = 2

for(i in 1:1000){
  x = rpois(n = n, lambda = 2)
  mean.val = sum(x) / n
  sqr.dev = (x-mean.val)^2

  var.n1[i] = sum(sqr.dev) / (n-1)
  var.n[i] = sum(sqr.dev) / n
}

# Compare estimators
bias.n1 = mean(var.n1 - var.true)
bias.n = mean(var.n - var.true)

precision.n1 = 1/var(var.n1)
precision.n = 1/var(var.n)

mse.n1 = bias.n1^2 + var(var.n1)
mse.n = bias.n^2 + var(var.n)

# Create numerical table
poisson.comparison = tibble(
  Estimator = c("Variance (n-1)", "Variance (n)"),
  Bias = c(bias.n1, bias.n),
  Precision = c(precision.n1,precision.n),
  MSE = c(mse.n1, mse.n)
)

poisson.comparison |>
  knitr::kable(digits = 2, caption = "Poisson Variance Estimator Comparison")
```

Table 1: Poisson Variance Estimator Comparison

Estimator	Bias	Precision	MSE
Variance (n-1)	-0.02	2.10	0.48
Variance (n)	-0.12	2.33	0.44

```r
# Create Plots
ggdat = tibble(
  value = c(var.n1, var.n),
  estimator = rep(c("Variance (n-1)", "Variance (n)"), each = 1000)
)

ggdat |> ggplot() +
  geom_density(aes(x=value, color=estimator))+
  geom_vline(xintercept = var.true, col = "red") +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Variance Estimate",
       y = "Density",
       color = "Estimator") +
  ggtitle("Distribution of Variance Estimators\n(Poisson, Lambda=2)")
```



Binomial:

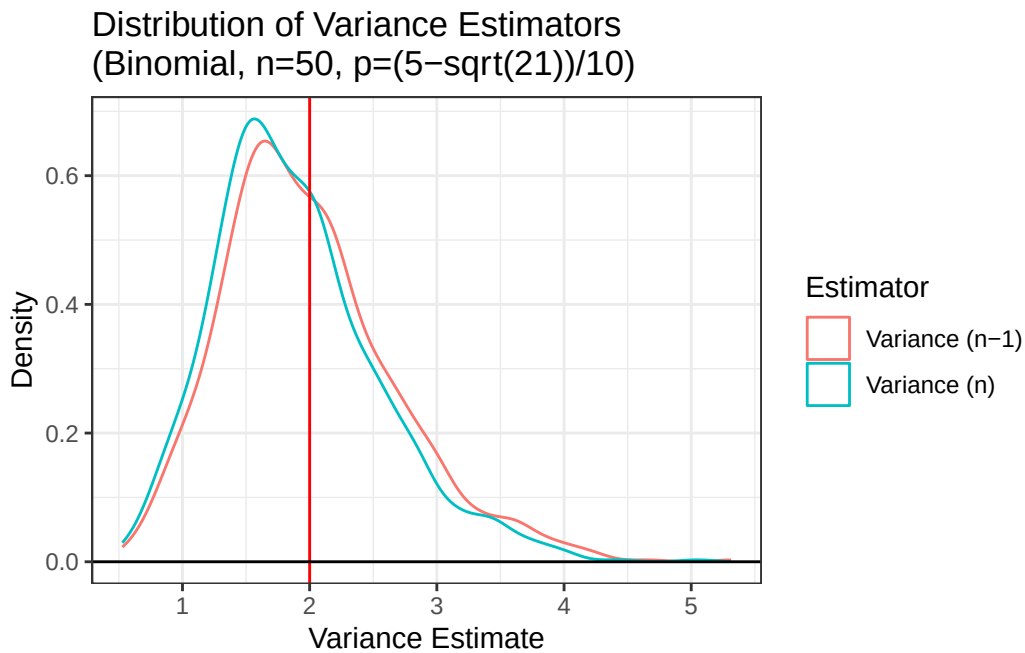
```r
set.seed(7272)

var.n1 = NA #initialized vector for S_{n-1}^2
var.n = NA #initialized vector for S_{n}^2
n = 20
var.true = 2

for(i in 1:1000){
  x = rbinom(n,50,(5-sqrt(21))/10)
  mean.val = sum(x) / n
  sqr.dev = (x-mean.val)^2

  var.n1[i] = sum(sqr.dev) / (n-1)
  var.n[i] = sum(sqr.dev) / n
}

# Compare estimators
```

```r
bias.n1 = mean(var.n1 - var.true)
bias.n = mean(var.n - var.true)

precision.n1 = 1/var(var.n1)
precision.n = 1/var(var.n)

mse.n1 = bias.n1^2 + var(var.n1)
mse.n = bias.n^2 + var(var.n)

# Create numerical table
binom.comparison = tibble(
  Estimator = c("Variance (n-1)", "Variance (n)"),
  Bias = c(bias.n1, bias.n),
  Precision = c(precision.n1,precision.n),
  MSE = c(mse.n1, mse.n)
)

binom.comparison |>
  knitr::kable(digits = 2, caption = "Binomial Variance Estimator Comparison")
```

Table 2: Binomial Variance Estimator Comparison

Estimator	Bias	Precision	MSE
Variance (n-1)	-0.01	2.13	0.47
Variance (n)	-0.11	2.36	0.44

```r
# Create Plots
ggdat = tibble(
  value = c(var.n1, var.n),
  estimator = rep(c("Variance (n-1)", "Variance (n)"), each = 1000)
)

ggdat |> ggplot() +
  geom_density(aes(x=value, color=estimator))+
  geom_vline(xintercept = var.true, col = "red") +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Variance Estimate",
       y = "Density",
       color = "Estimator") +
  ggtitle("Distribution of Variance Estimators\n(Binomial, n=50, p=(5-sqrt(21))/10)")
```



Normal:

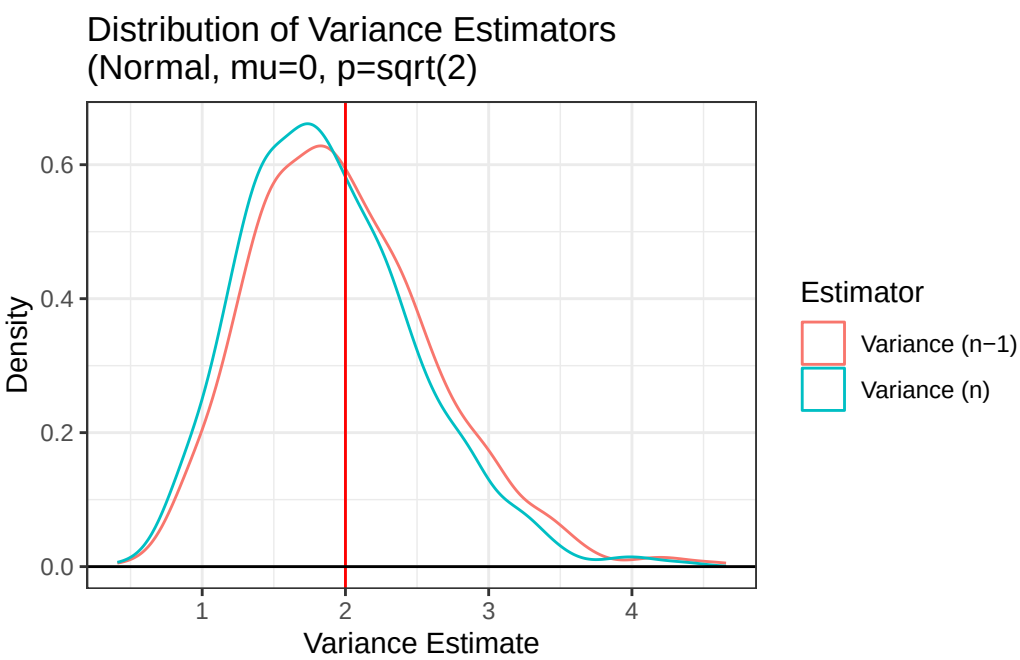
```r
set.seed(7272)

var.n1 = NA #initialized vector for S_{n-1}^2
var.n = NA #initialized vector for S_{n}^2
n = 20
var.true = 2

for(i in 1:1000){
  x = rnorm(n,0,sqrt(2))
  mean.val = sum(x) / n
  sqr.dev = (x-mean.val)^2

  var.n1[i] = sum(sqr.dev) / (n-1)
  var.n[i] = sum(sqr.dev) / n
}

# Compare estimators
bias.n1 = mean(var.n1 - var.true)
bias.n = mean(var.n - var.true)

precision.n1 = 1/var(var.n1)
precision.n = 1/var(var.n)

mse.n1 = bias.n1^2 + var(var.n1)
mse.n = bias.n^2 + var(var.n)

# Create numerical table
norm.comparison = tibble(
  Estimator = c("Variance (n-1)", "Variance (n)"),
  Bias = c(bias.n1, bias.n),
  Precision = c(precision.n1,precision.n),
  MSE = c(mse.n1, mse.n)
)

```

```r
norm.comparison |>
  knitr::kable(digits = 2, caption = "Normal Dist. Variance Estimator Comparison")
```

Table 3: Normal Dist. Variance Estimator Comparison

Estimator	Bias	Precision	MSE
Variance (n-1)	-0.01	2.41	0.41
Variance (n)	-0.11	2.67	0.39

```r
# Create Plots
ggdat = tibble(
  value = c(var.n1, var.n),
  estimator = rep(c("Variance (n-1)", "Variance (n)"), each = 1000)
)

ggdat |> ggplot() +
  geom_density(aes(x=value, color=estimator))+
  geom_vline(xintercept = var.true, col = "red") +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Variance Estimate",
       y = "Density",
       color = "Estimator") +
  ggtitle("Distribution of Variance Estimators\n(Normal, mu=0, p=sqrt(2)")
```



Exponential:

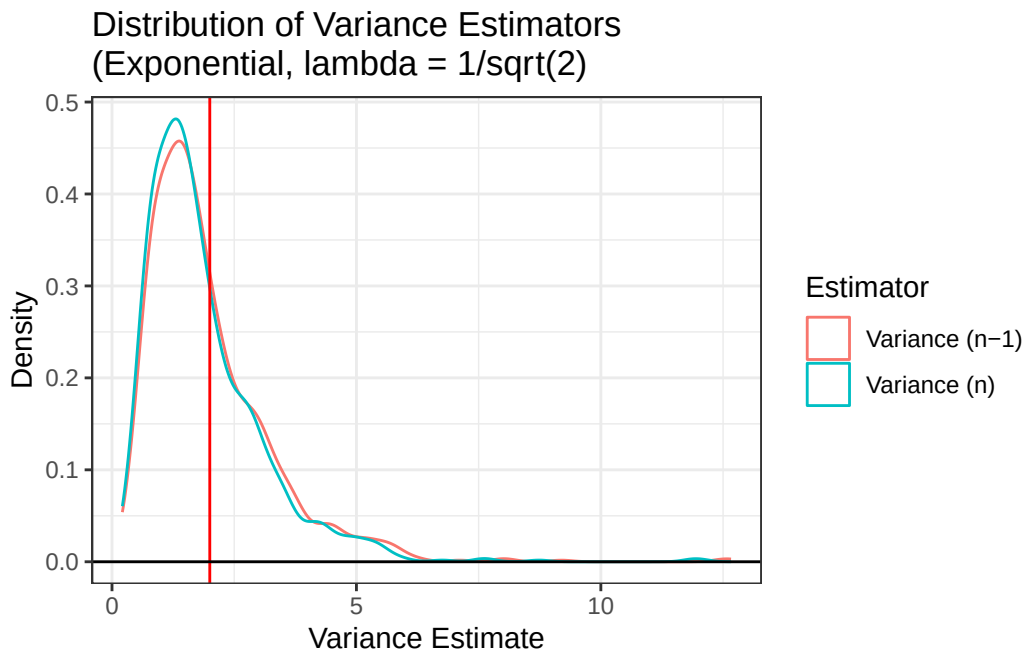
```r
set.seed(7272)

var.n1 = NA #initialized vector for S_{n-1}^2
var.n = NA #initialized vector for S_{n}^2
n = 20
var.true = 2

```

```r
for(i in 1:1000){
  x = rexp(n,rate=1/sqrt(2))
  mean.val = sum(x) / n
  sqr.dev = (x-mean.val)^2

  var.n1[i] = sum(sqr.dev) / (n-1)
  var.n[i] = sum(sqr.dev) / n
}

# Compare estimators
bias.n1 = mean(var.n1 - var.true)
bias.n = mean(var.n - var.true)

precision.n1 = 1/var(var.n1)
precision.n = 1/var(var.n)

mse.n1 = bias.n1^2 + var(var.n1)
mse.n = bias.n^2 + var(var.n)

# Create numerical table
exp.comparison = tibble(
  Estimator = c("Variance (n-1)", "Variance (n)"),
  Bias = c(bias.n1, bias.n),
  Precision = c(precision.n1,precision.n),
  MSE = c(mse.n1, mse.n)
)

exp.comparison |>
  knitr::kable(digits = 2, caption = "Exp. Distr. Variance Estimator Comparison")
```

Table 4: Exp. Distr. Variance Estimator Comparison

Estimator	Bias	Precision	MSE
Variance (n-1)	-0.03	0.61	1.64
Variance (n)	-0.12	0.68	1.49

```r
# Create Plots
ggdat = tibble(
  value = c(var.n1, var.n),
  estimator = rep(c("Variance (n-1)", "Variance (n)"), each = 1000)
)

ggdat |> ggplot() +
  geom_density(aes(x=value, color=estimator))+
  geom_vline(xintercept = var.true, col = "red") +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Variance Estimate",
       y = "Density",
       color = "Estimator") +
  ggtitle("Distribution of Variance Estimators\n(Exponential, lambda = 1/sqrt(2)")
```



2.2 Simulation over various sample sizes

Consider the effect of the sample size have on these results. Provide a graphical and numerical summary of what happens as n increases – say $n = 20$, $n = 50$, $n = 100$, and $n = 500$. Does the result in your initial simulation hold across samples? Or, do things change as the sample size changes? Ensure to set your randomization seed.

Solution:
Note: I used `echo=FALSE` to make my PDF easier to look at whilst working on it (I took a "brute force" approach to this section). Apologies if this is an issue.

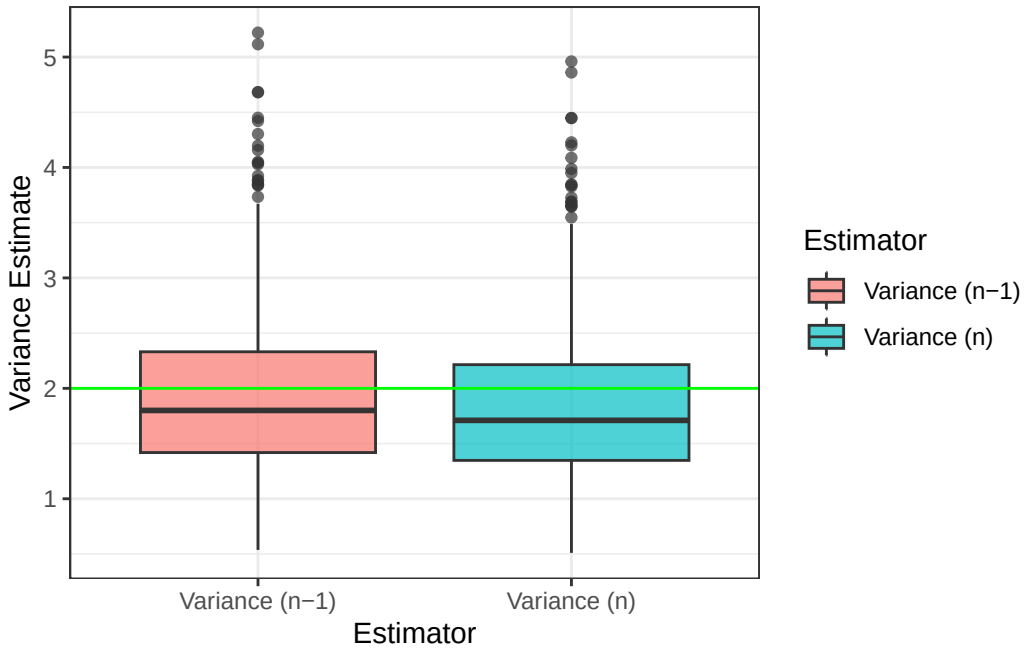
Generalizing the previous work into a function (code hidden):

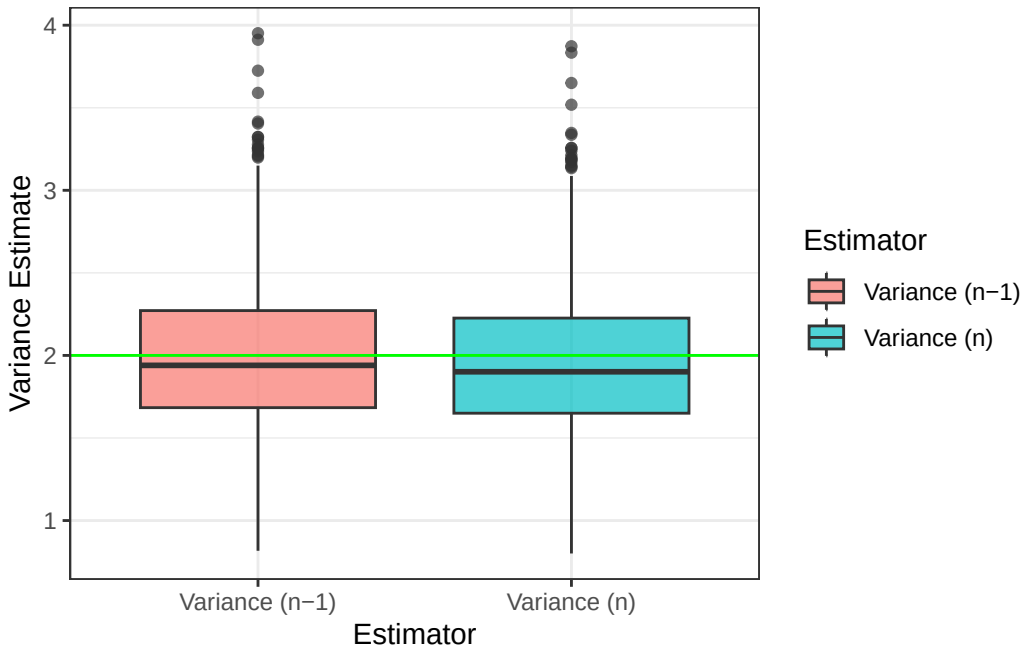
Running the Simulations (code hidden): Apologies for the looooong brute force coding below. This was just the most intuitive way for me to generate everything!

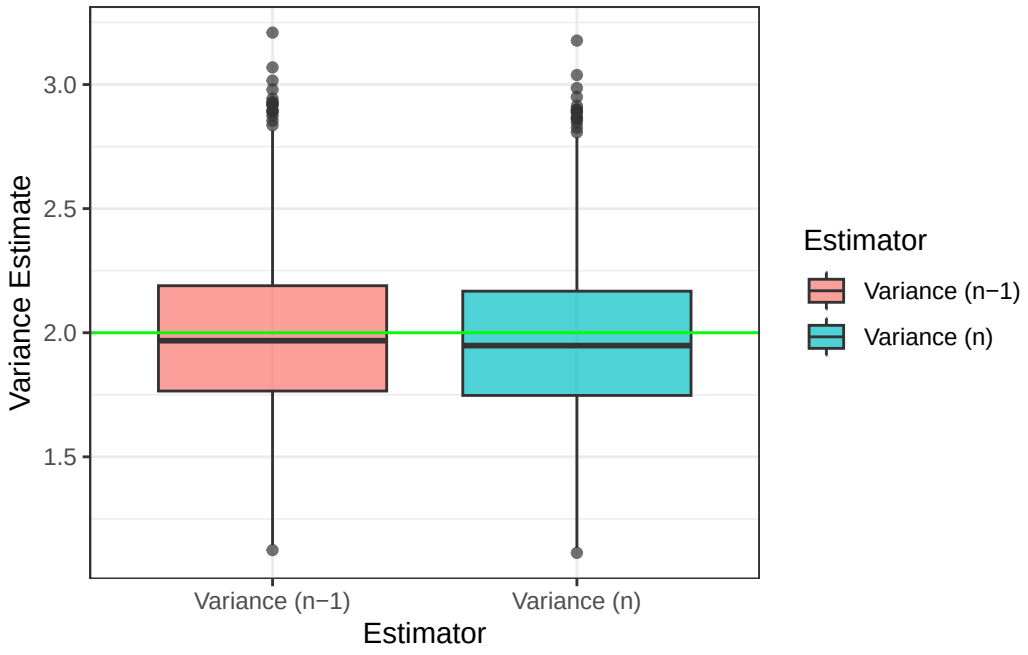
Poisson

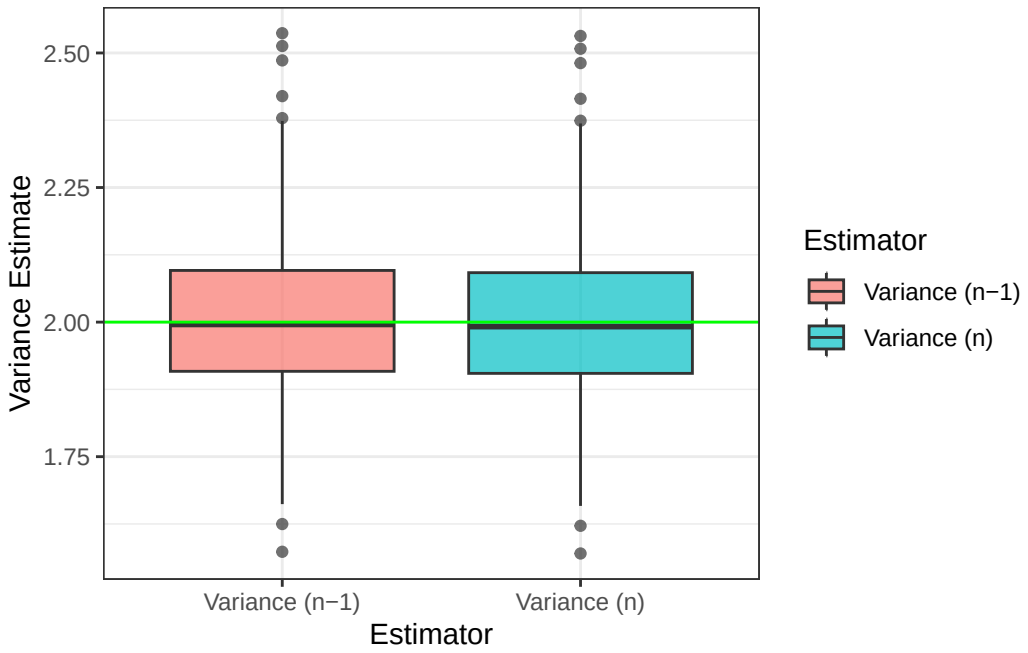
Table 5: Poisson Variance Estimators Across Sample Sizes

n	Bias (n-1)	Bias (n)	Precision (n-1)	Precision (n)	MSE (n-1)	MSE (n)
20	-0.0767	-0.1729	2.0849	2.3102	0.4855	0.4628
50	-0.0101	-0.0499	5.0170	5.2239	0.1994	0.1939
100	-0.0117	-0.0316	9.6444	9.8403	0.1038	0.1026
500	0.0050	0.0010	50.5581	50.7609	0.0198	0.0197


Variance Estimate
5
4
3
2
1
Variance (n−1)
Variance (n)
Estimator
Estimator
Variance (n−1)
Variance (n)


Variance Estimate
4
3
2
1
Variance (n−1)
Variance (n)
Estimator
Estimator
Variance (n−1)
Variance (n)

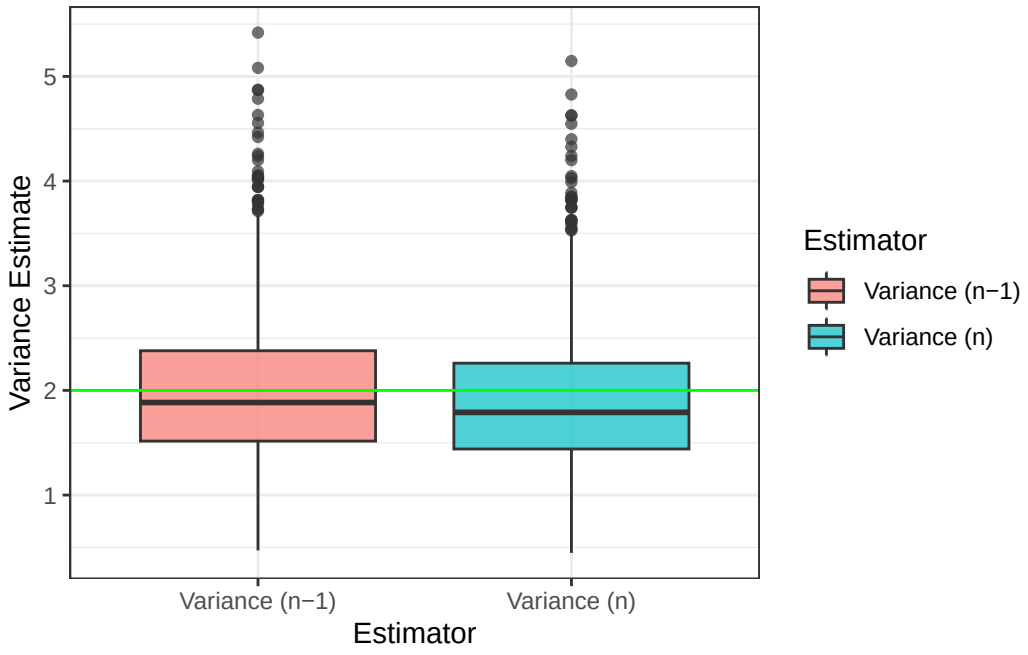


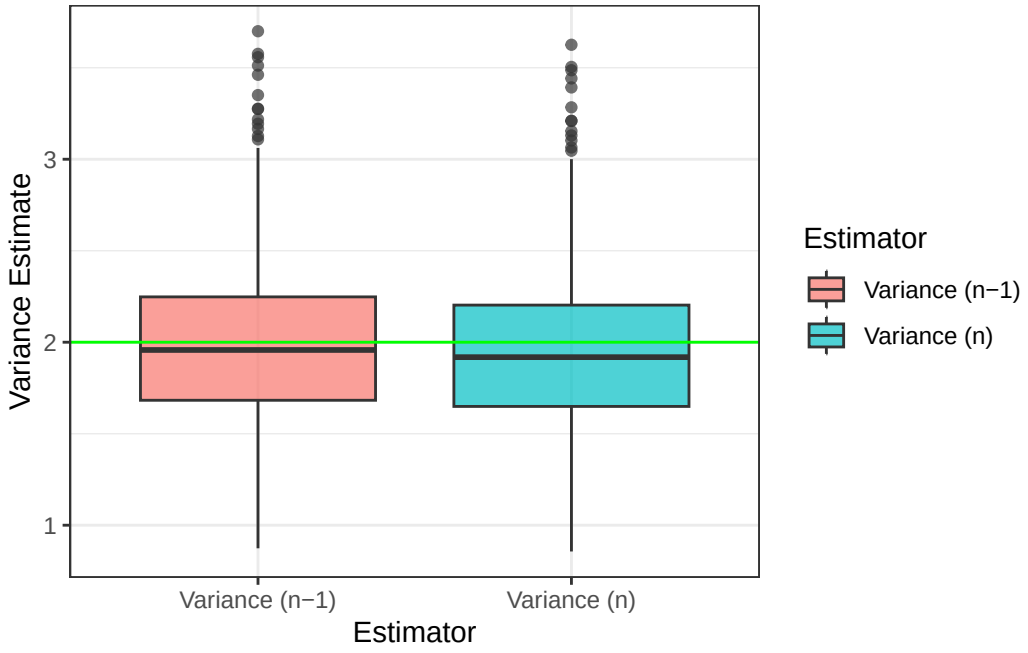


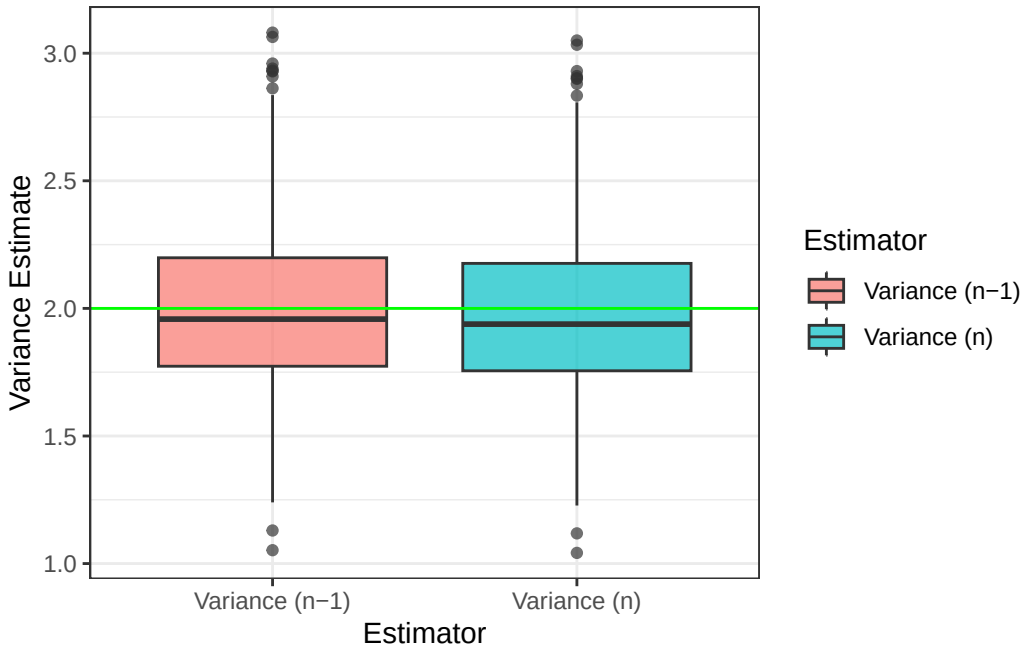
Binomial

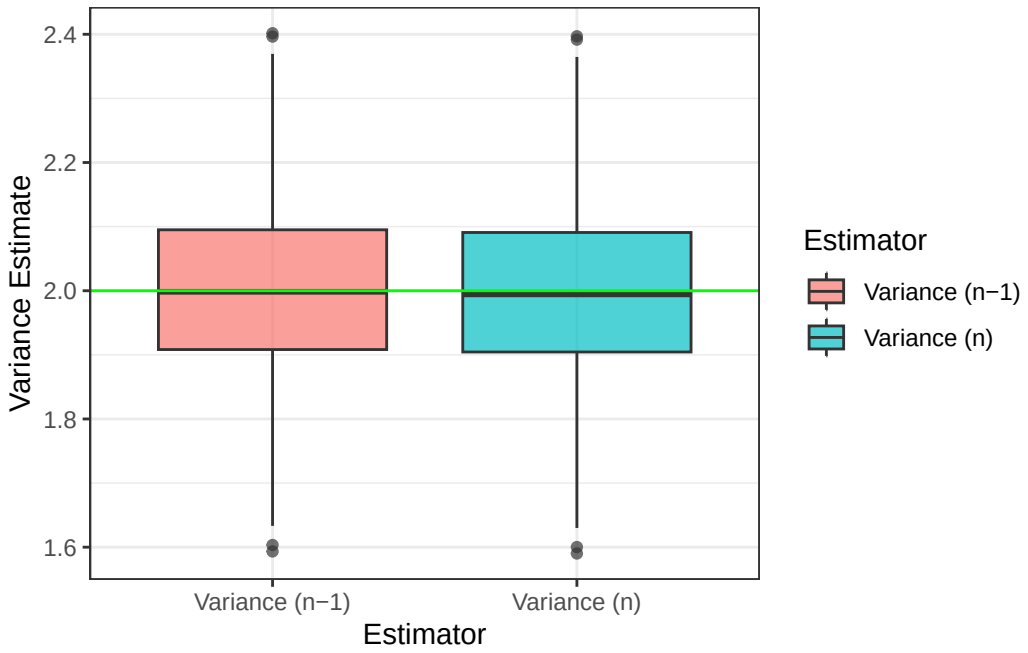
Table 6: Binomial Variance Estimators Across Sample Sizes

n	Bias (n-1)	Bias (n)	Precision (n-1)	Precision (n)	MSE (n-1)	MSE (n)
20	0.0043	-0.0959	1.8697	2.0717	0.5349	0.4919
50	-0.0075	-0.0473	5.3049	5.5236	0.1886	0.1833
100	-0.0107	-0.0306	9.6881	9.8848	0.1033	0.1021
500	0.0025	-0.0015	53.1977	53.4111	0.0188	0.0187


Variance Estimate
5
4
3
2
1
Variance (n−1)
Variance (n)
Estimator
Estimator
Variance (n−1)
Variance (n)


Variance Estimate
3
2
1
Variance (n−1)
Variance (n)
Estimator
Estimator
Variance (n−1)
Variance (n)

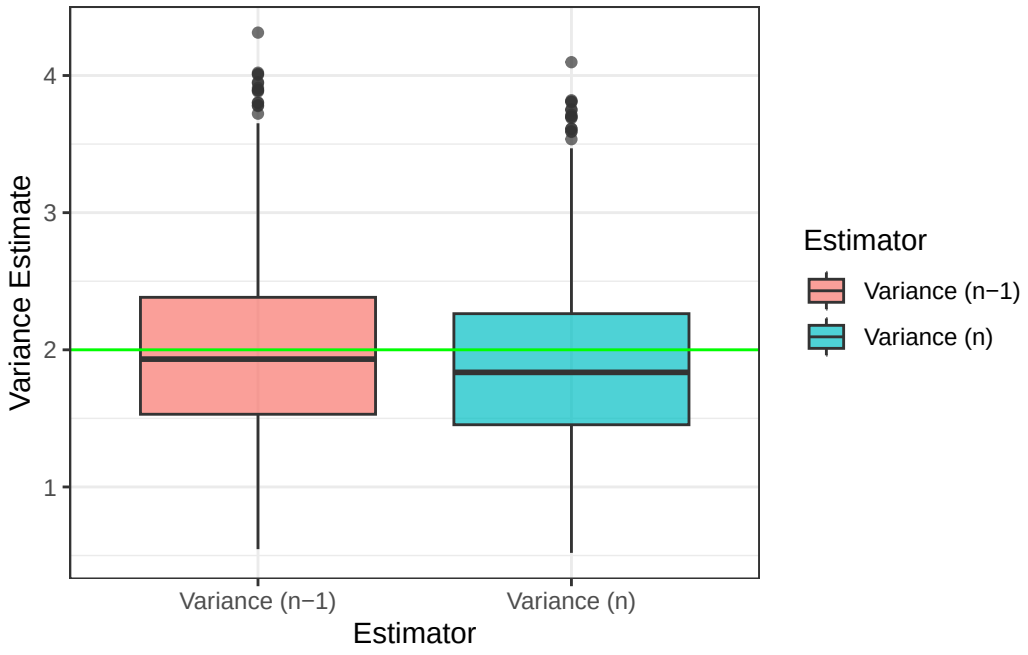


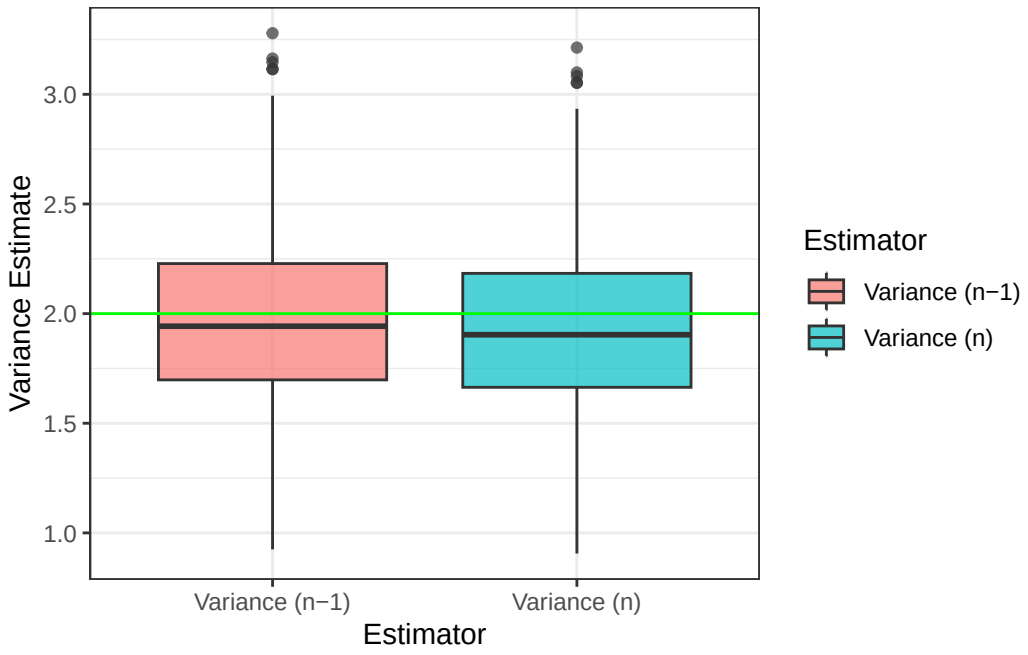


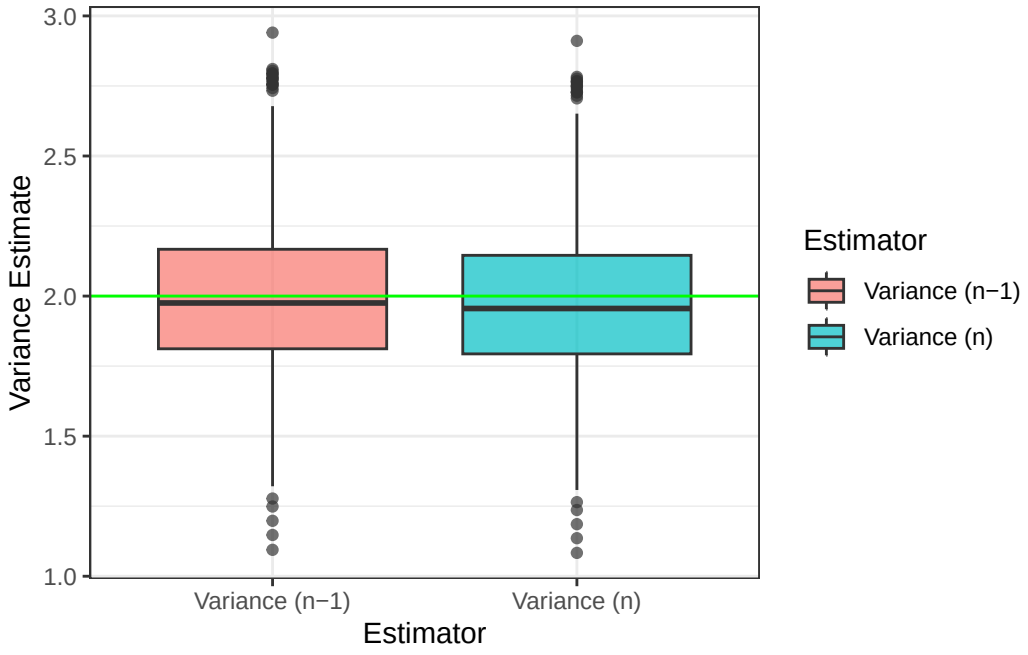
Normal

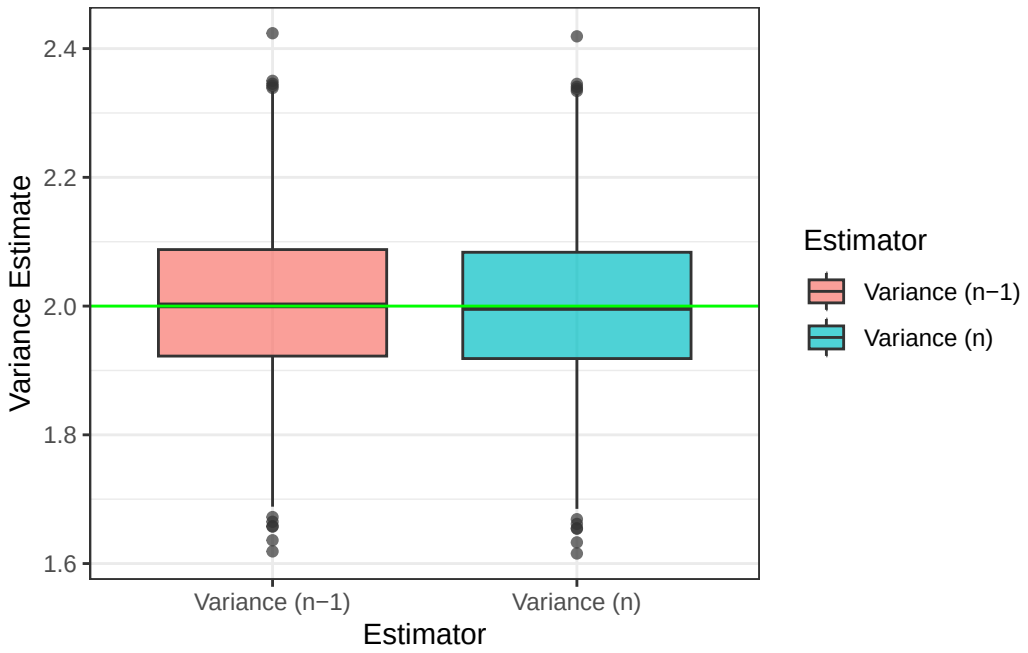
Table 7: Normal Distribution Variance Estimators Across Sample Sizes

n	Bias (n-1)	Bias (n)	Precision (n-1)	Precision (n)	MSE (n-1)	MSE (n)
20	0.0008	-0.0992	2.5488	2.8241	0.3923	0.3639
50	-0.0303	-0.0697	6.3219	6.5826	0.1591	0.1568
100	-0.0087	-0.0286	12.3458	12.5965	0.0811	0.0802
500	0.0051	0.0011	62.3261	62.5761	0.0161	0.0160


Variance Estimate
4
3
2
1
Variance (n−1)
Variance (n)
Estimator
Estimator
Variance (n−1)
Variance (n)


Variance Estimate
3.0
2.5
2.0
1.5
1.0
Variance (n−1)
Variance (n)
Estimator
Estimator
Variance (n−1)
Variance (n)

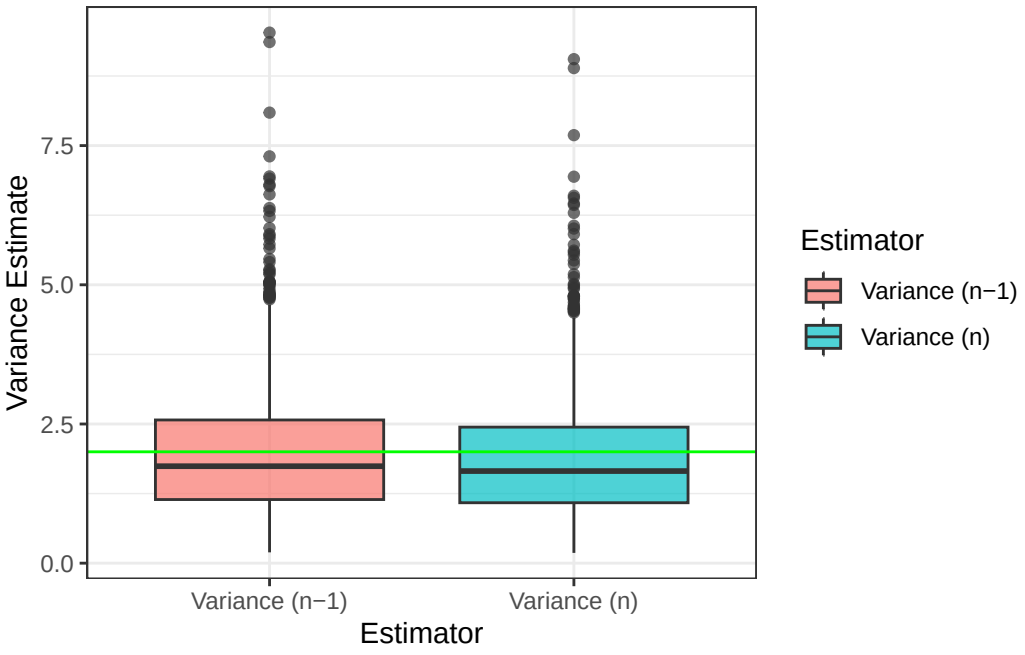


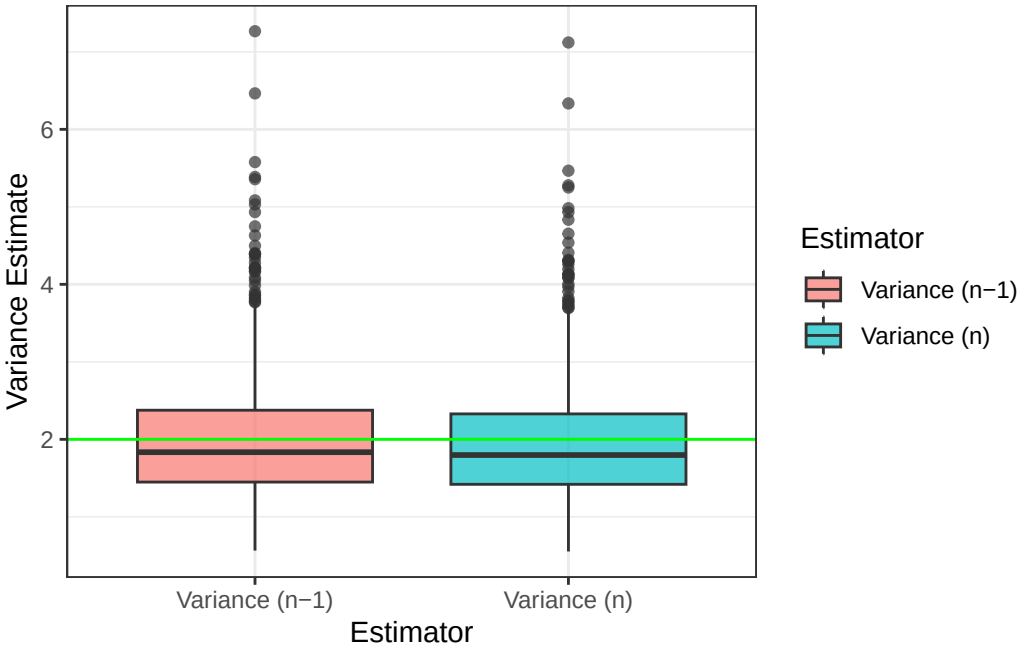


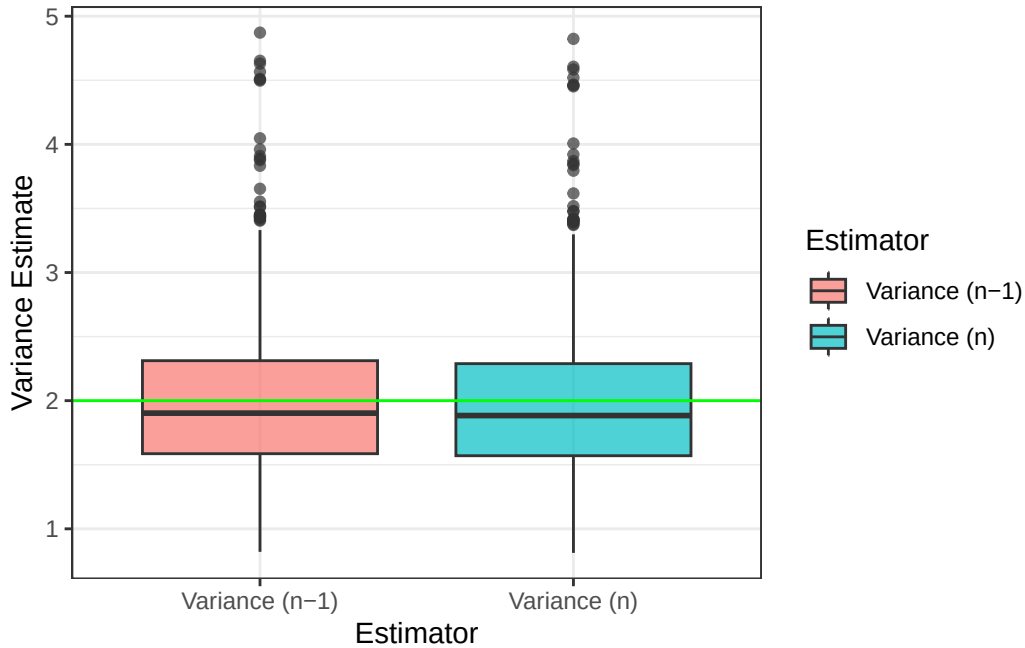
Exponential

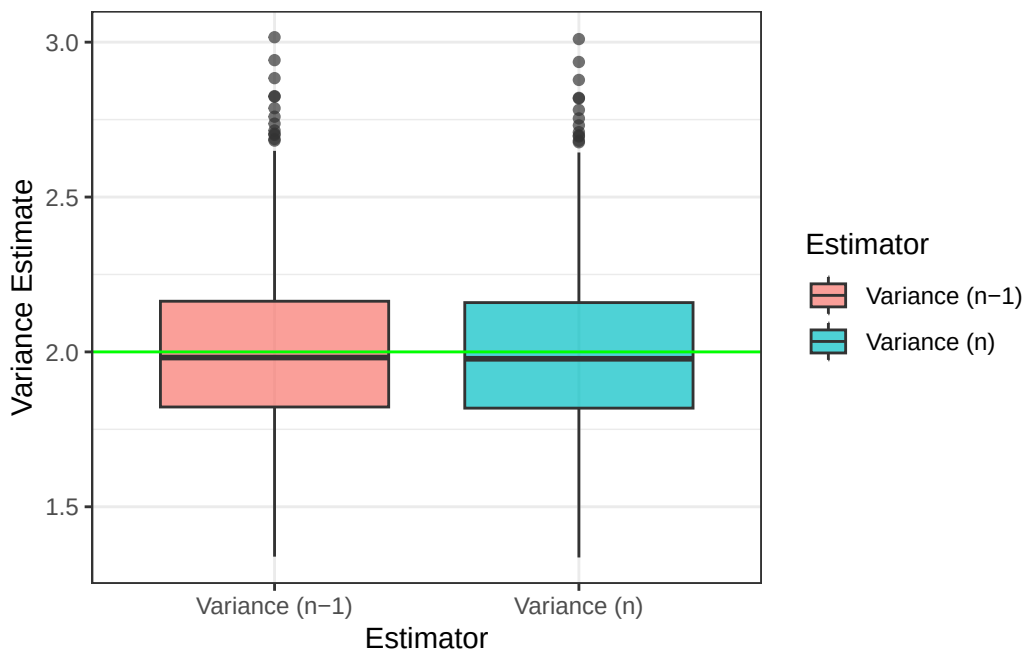
Table 8: Exponential Distribution Variance Estimators Across Sample Sizes

n	Bias (n-1)	Bias (n)	Precision (n-1)	Precision (n)	MSE (n-1)	MSE (n)
20	0.0185	-0.0824	0.6449	0.7145	1.5511	1.4063
50	-0.0066	-0.0465	1.5282	1.5912	0.6544	0.6306
100	-0.0060	-0.0260	2.9505	3.0104	0.3390	0.3329
500	0.0022	-0.0018	15.5934	15.6560	0.0641	0.0639


Variance Estimate
7.5
5.0
2.5
0.0
Variance (n−1)
Variance (n)
Estimator
Estimator
Variance (n−1)
Variance (n)


Variance Estimate
6
4
2
Variance (n−1)
Variance (n)
Estimator
Estimator
Variance (n−1)
Variance (n)



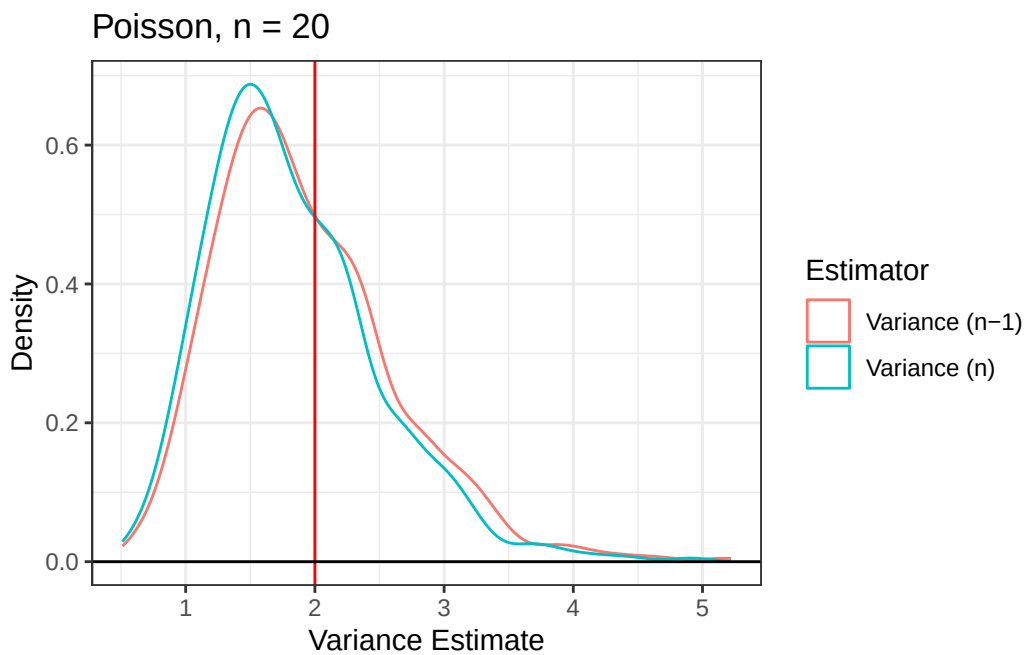


Commentary:
Across all four of the distributions, we find that the results change with sample size. Specifically, as n grows,the two density plots for each distribution (one for each variance measure) converge, and it becomes harder to differentiate between the density plot for s^2_{n-1} and the density plot for s^2_n. This phenomenon is further supported by the numerical summary tables. As n increases, we notice that bias and MSE for both s^2_{n-1} and s^2_n approach 0, while precision increases for both.

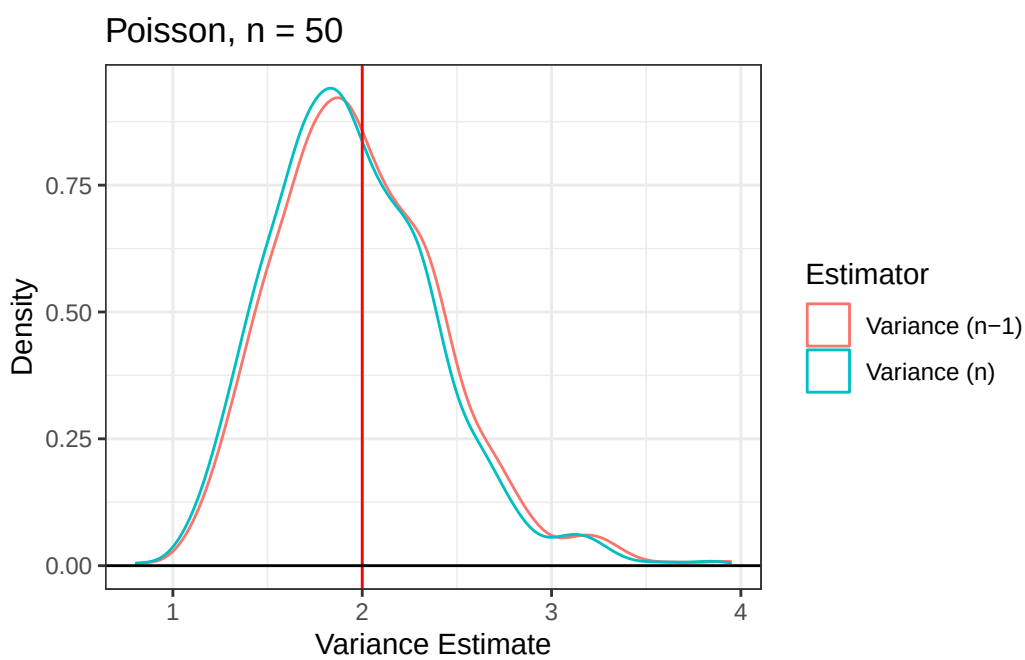
2.3 Simulation

You have written code to draw a random sample of various sizes from each distribution, and to calculate and store the calculated estimates. Rework your code to produce a density plot of the s^2_{n-1} and s^2_n. Do you have a guess about their distribution? Ensure to set your randomization seed.

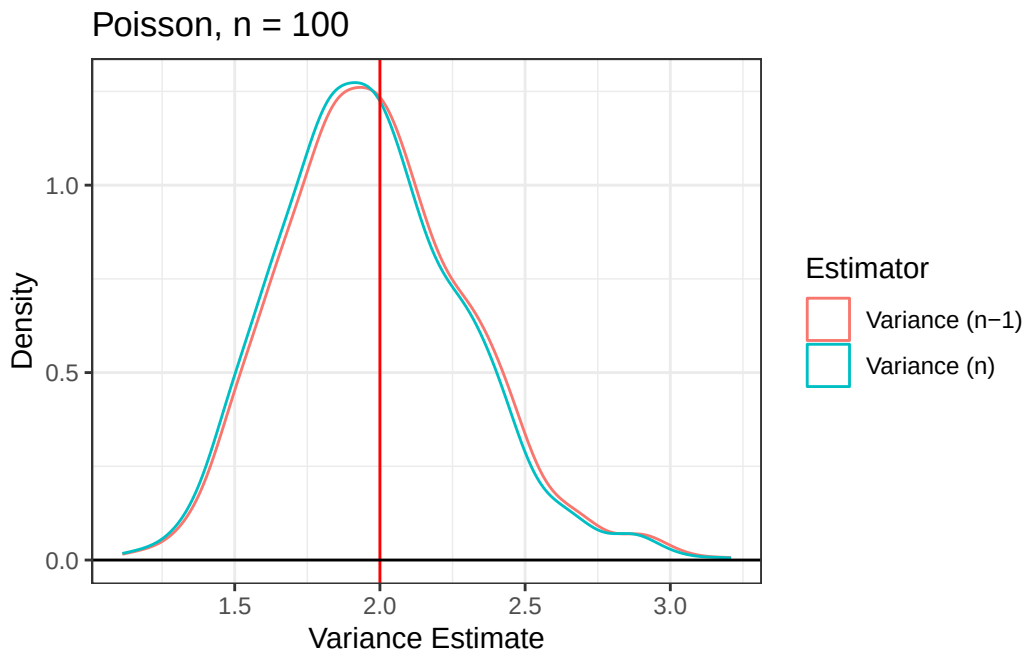
```r
pois20$density.plot
```



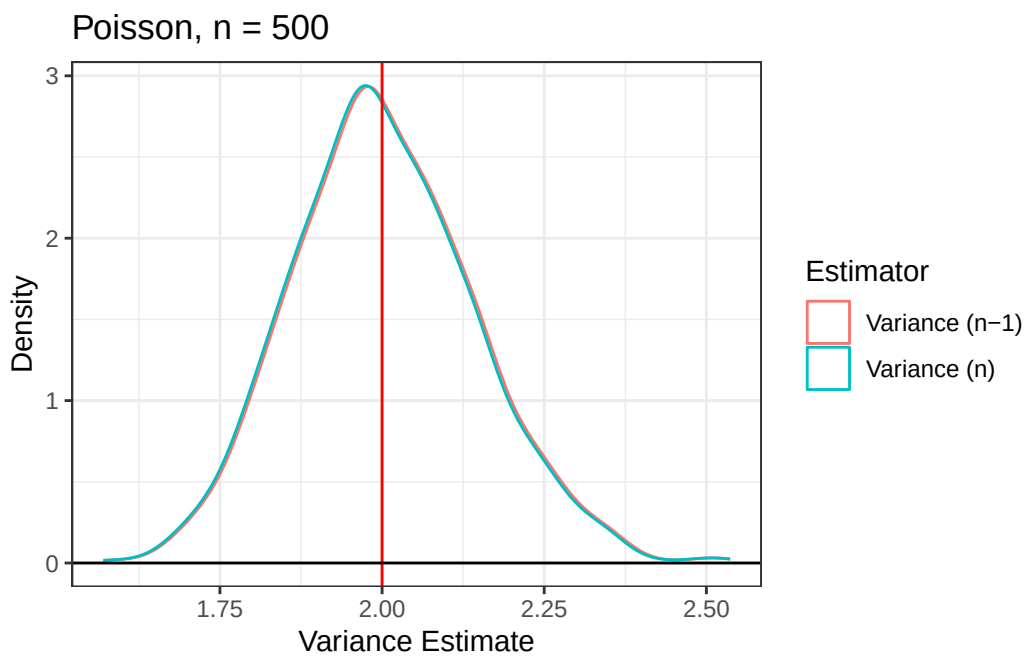

```r
pois50$density.plot
```



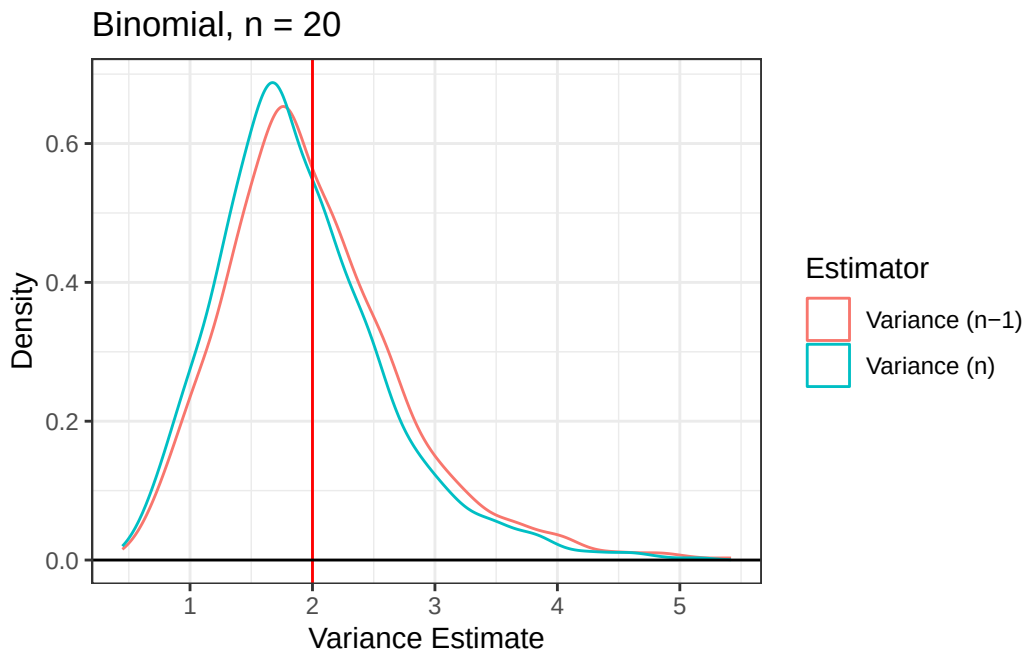

```r
pois100$density.plot
```



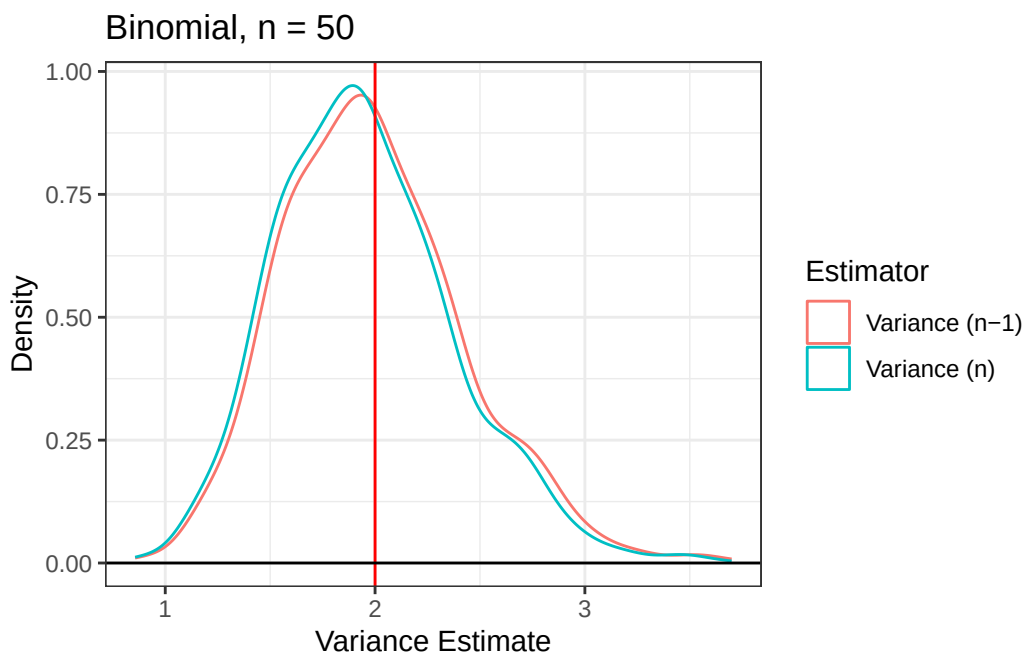

```
pois500$density.plot
```



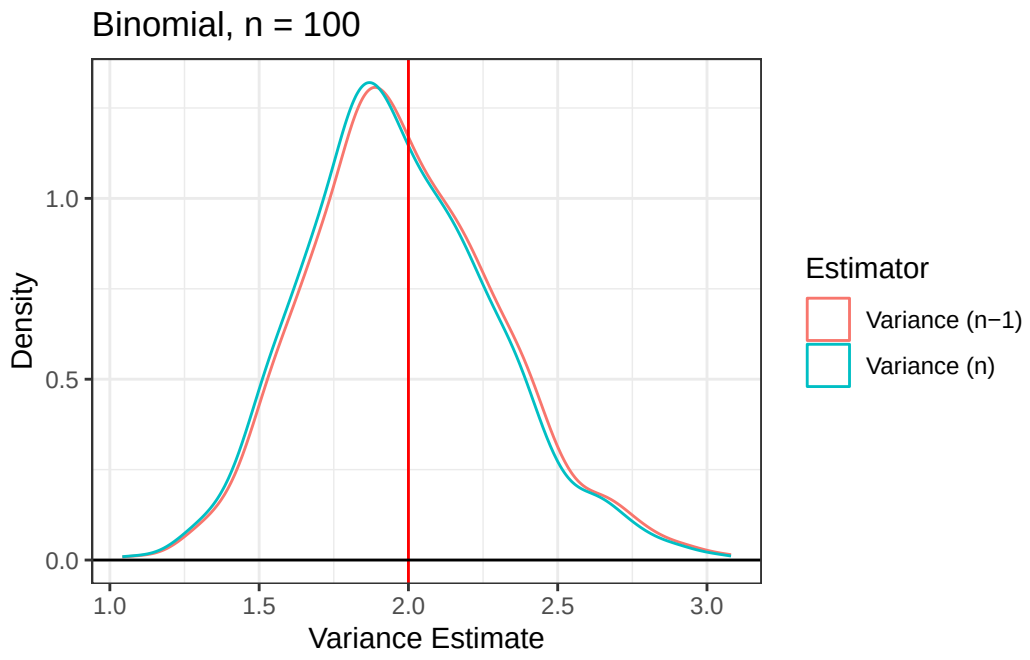

```
binom20$density.plot
```



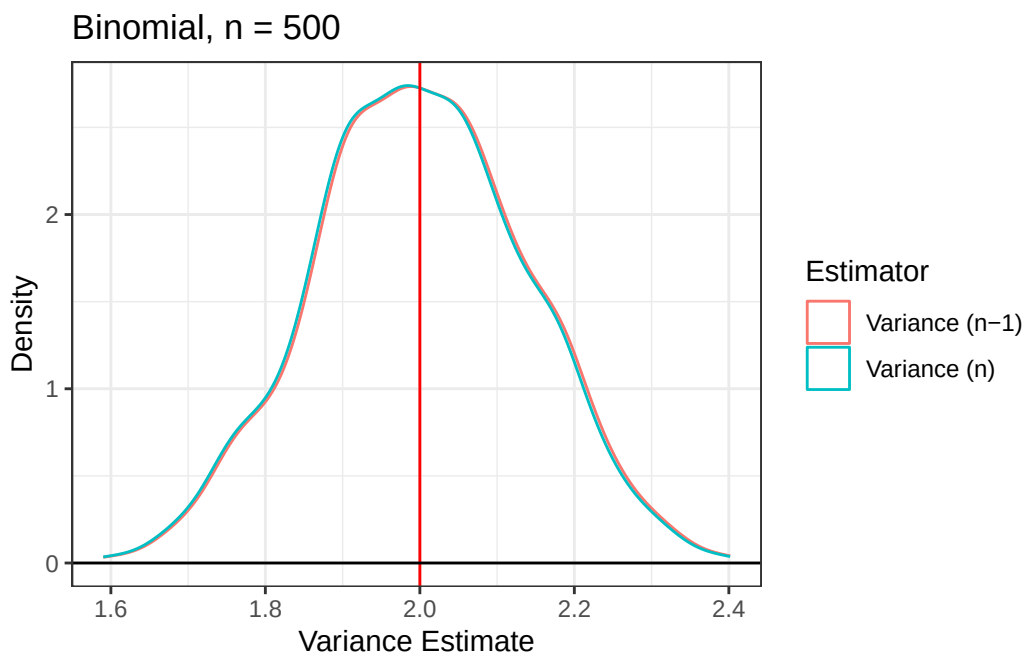

```
binom50$density.plot
```



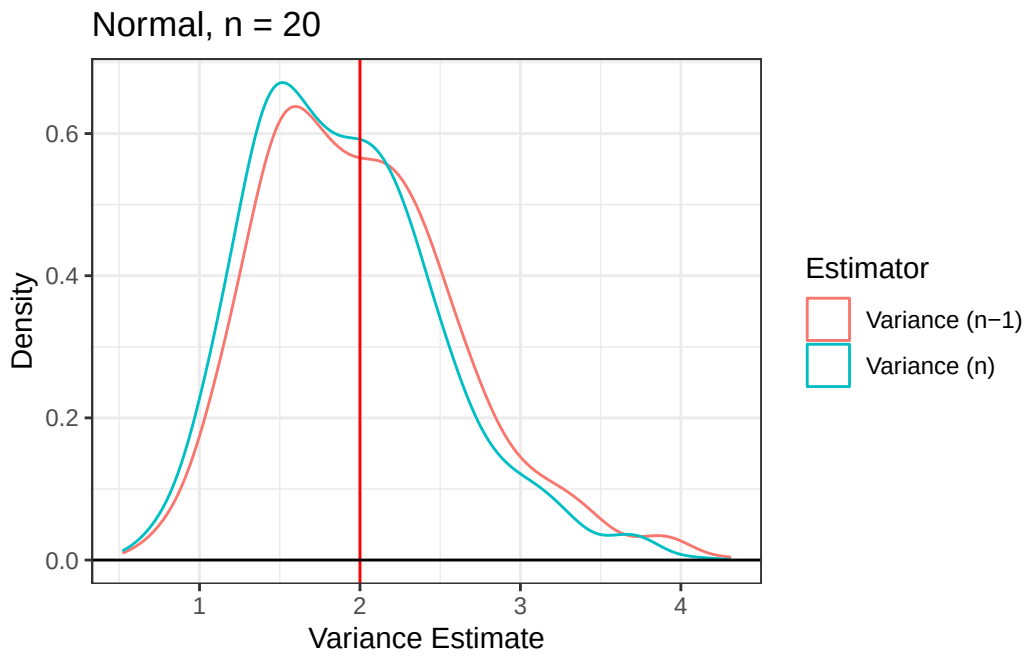

```
binom100$density.plot
```



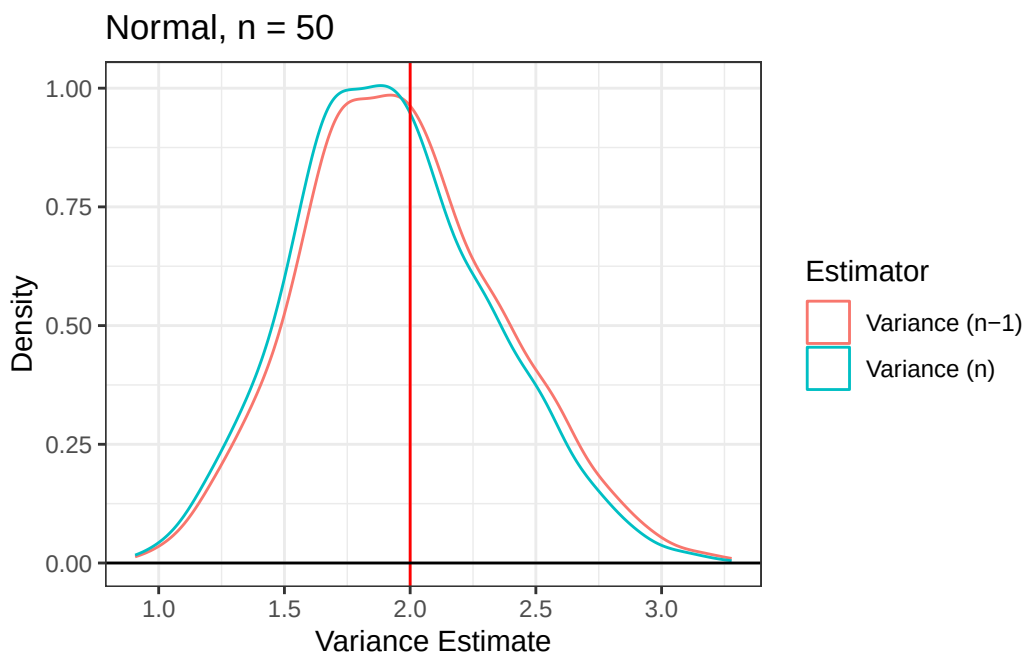

```
binom500$density.plot
```



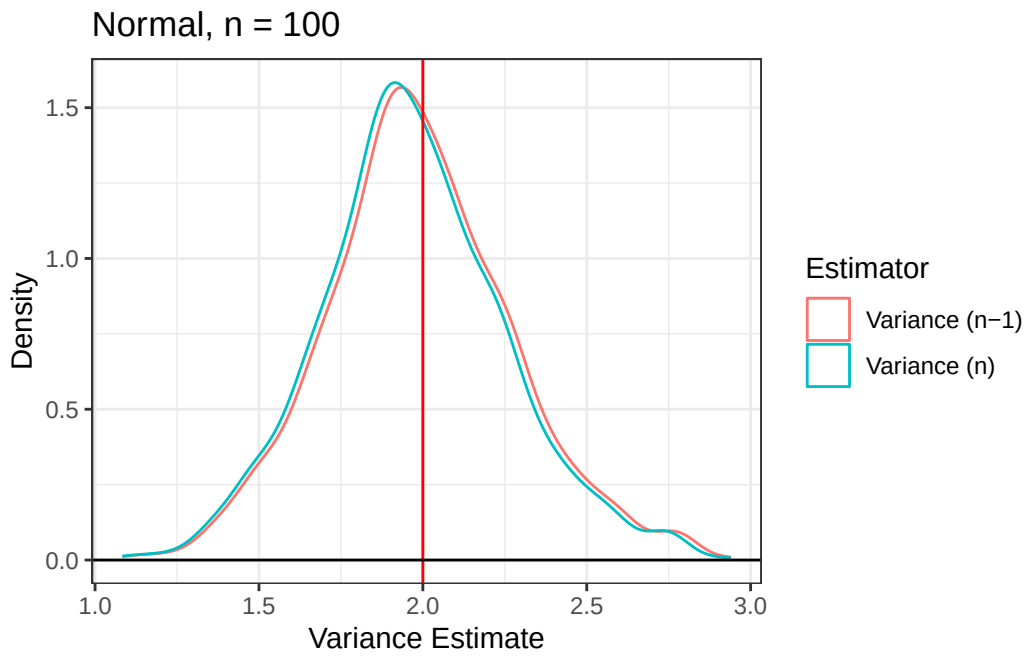

```
norm20$density.plot
```



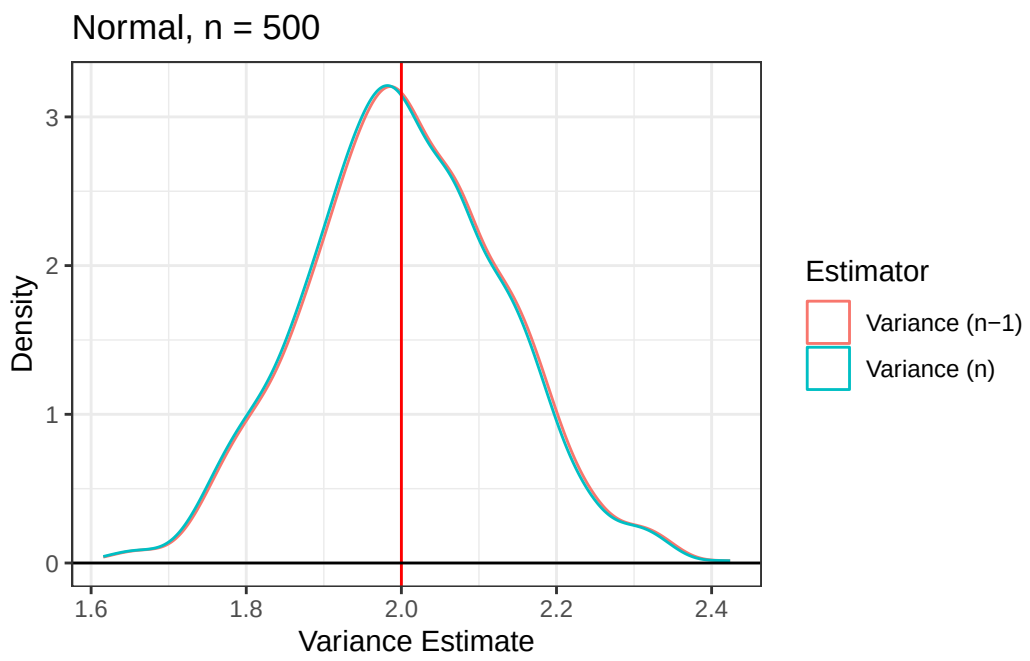

```
norm50$density.plot
```



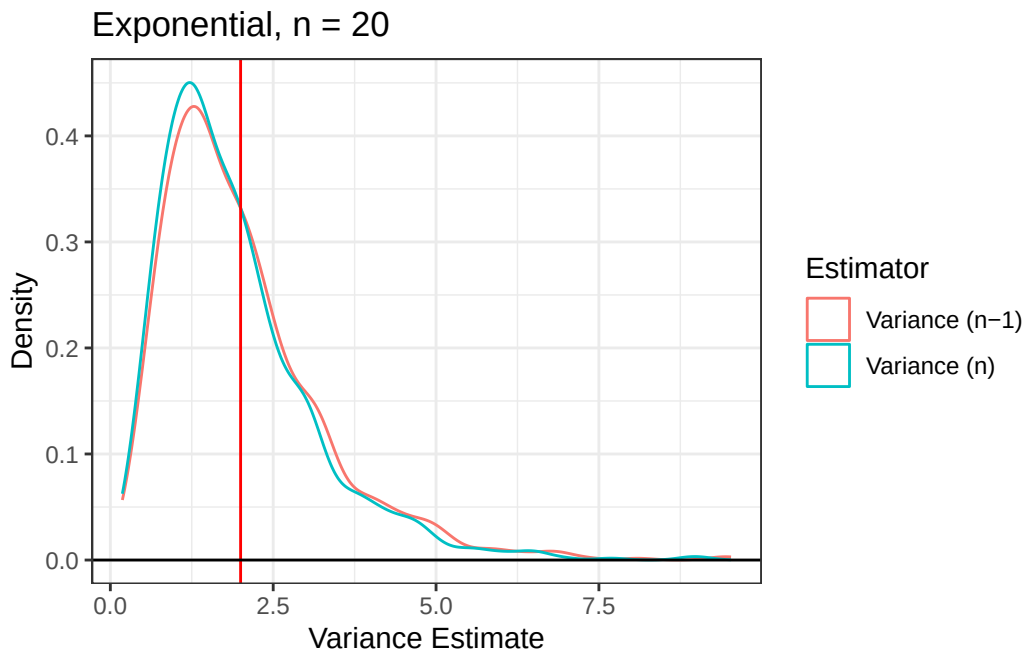

```
norm100$density.plot
```



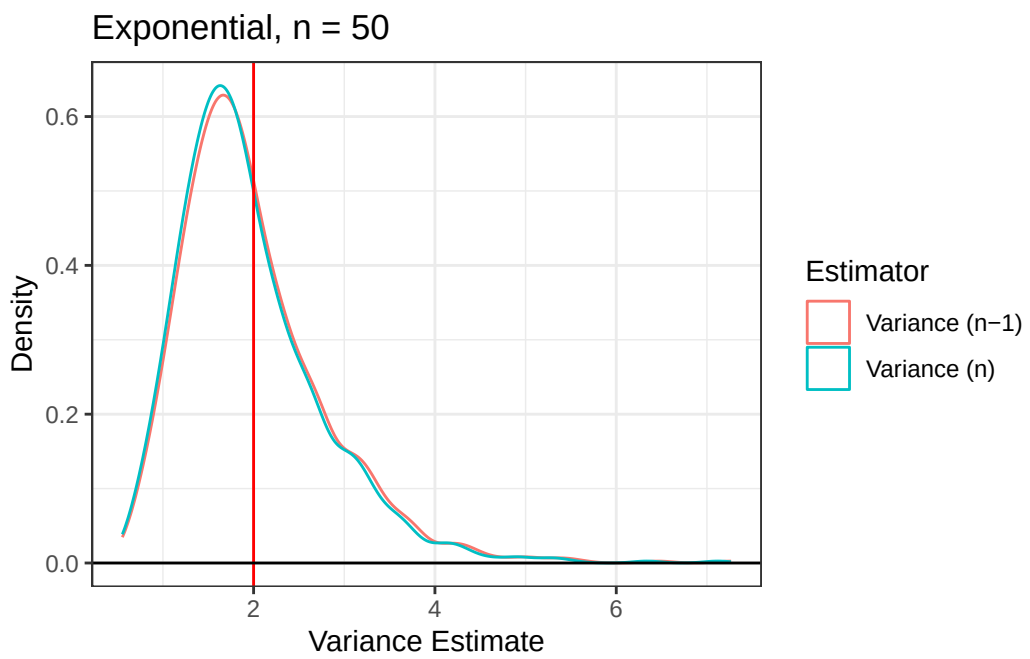

```
norm500$density.plot
```



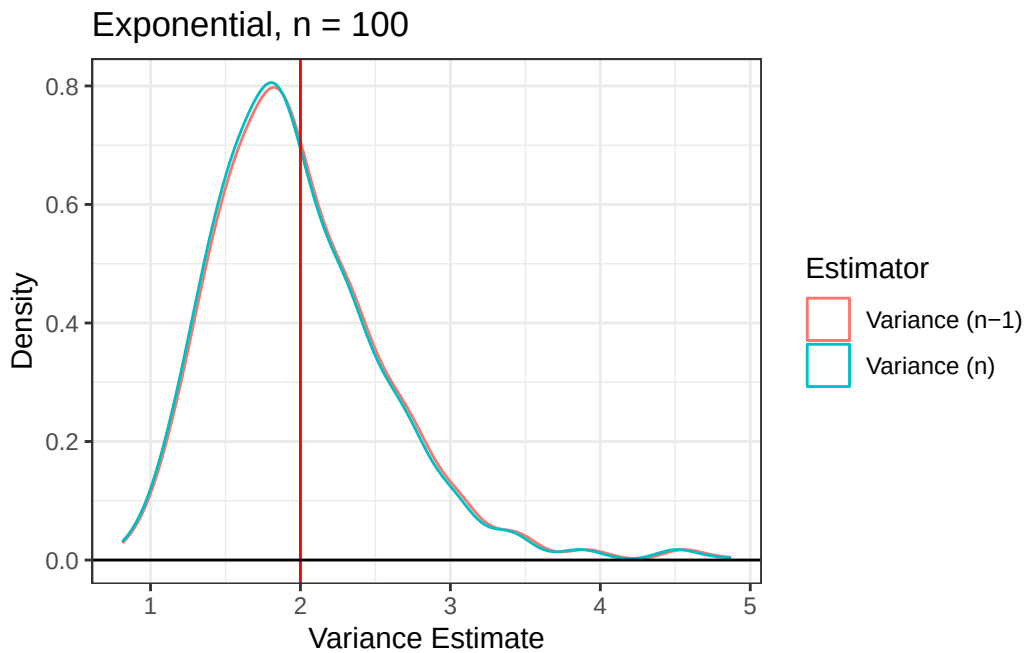

```
exp20$density.plot
```



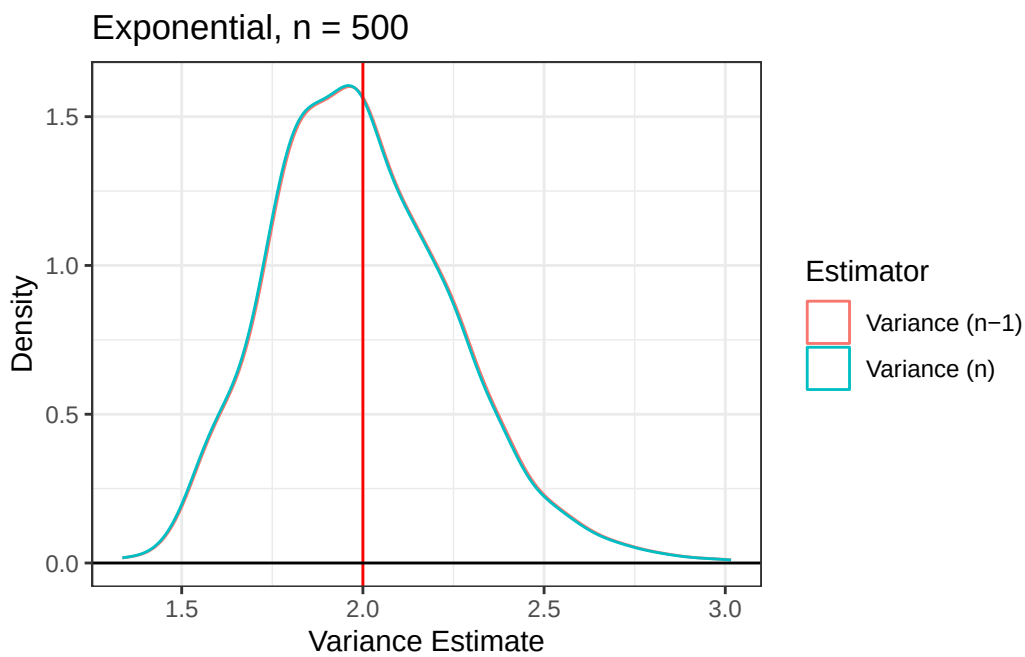

```
exp50$density.plot
```




```
exp100$density.plot
```




```
exp500$density.plot
```



Commentary Density plots for the four distributions with $n = 100$, we can obtain a good estimate about the distributions for s^2_{n-1} and s^2_n. The first thing the density plots demonstrate is that s^2_{n-1} and s^2_n have very similar distributions (despite s^2_n being slightly biased toward the left), though what distribution this is depends on the distribution of the model, and the sample size, n.

Small n: For small n, the distributions for s^2_{n-1} and s^2_n in all distributions are right-skewed.

Large n: With large enough $n(= 500)$, s^2_{n-1} and s^2_n for the Poisson, Binomial, Normal, and Exponential distributions, all appear to approach a symmetric distribution. However, for the Exponential distribution, s^2_{n-1} and s^2_n still appears to follow a noticeably right-skewed distribution even at $n = 100$, taking a bit longer than the others to approach this symmetric shape.

The graphs also illustrate the reduced variance/improved precision as n increases, with observations further from

the mean becoming less likely as n increases.

2.4 Bootstrap

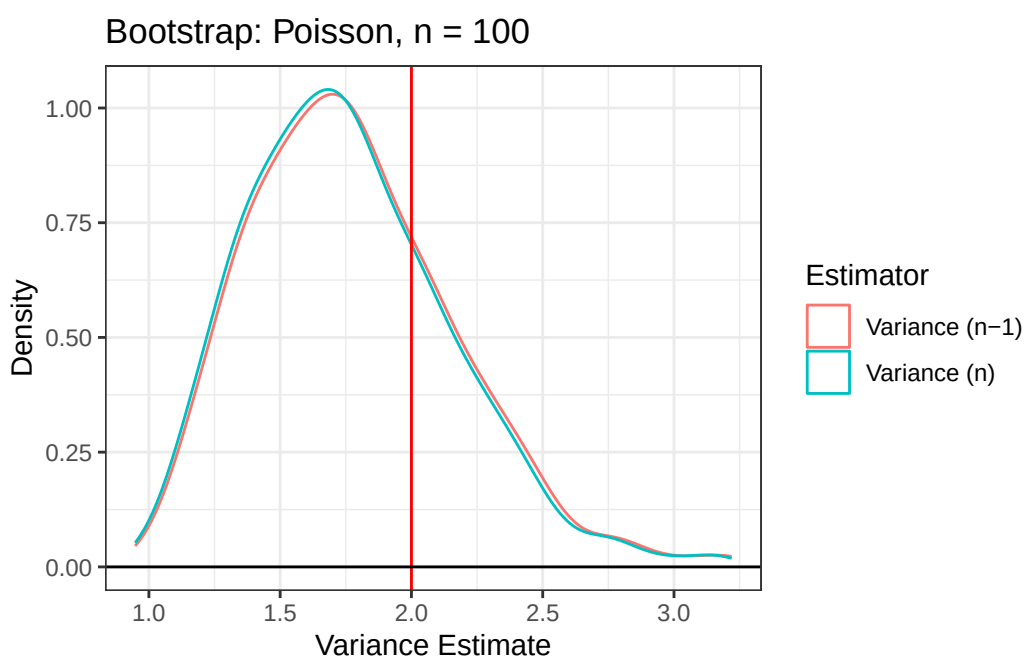
Now, take *one* sample from each distribution. Instead of drawing a random sample at each iteration, sample *with replacement* from the one sample. Produce a density plot of the s^2_{n-1} and s^2_n. Do we get roughly the same information as we do in the full simulation? Try this over several seeds – what do you notice? Ensure to set your randomization seed.

NOTE:
Turn off `#|echo=FALSE` to see my clunky code.

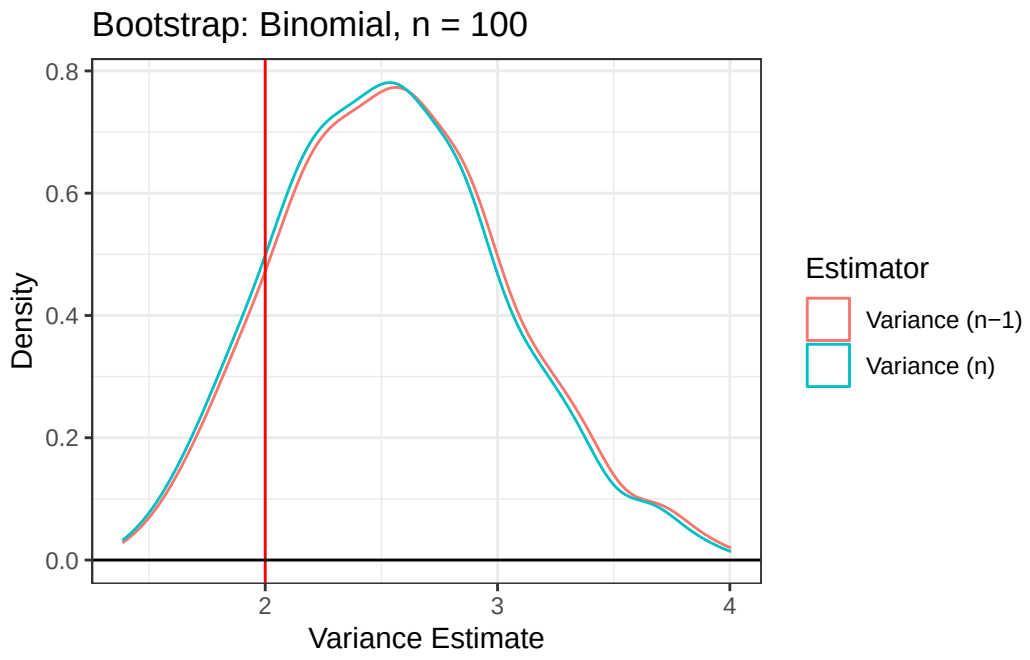
Table 9: Bootstrap Variance Estimators Across Distributions

Distribution	Bias (n-1)	Bias (n)	Precision (n-1)	Precision (n)	MSE (n-1)	MSE (n)
Poisson	-0.2246	-0.2423	6.4919	6.6237	0.2045	0.2097
Binomial	0.5639	0.5383	4.1699	4.2546	0.5578	0.5248
Normal	0.1628	0.1412	11.7911	12.0305	0.1113	0.1031
Exponential	-0.4963	-0.5113	9.4475	9.6393	0.3521	0.3652

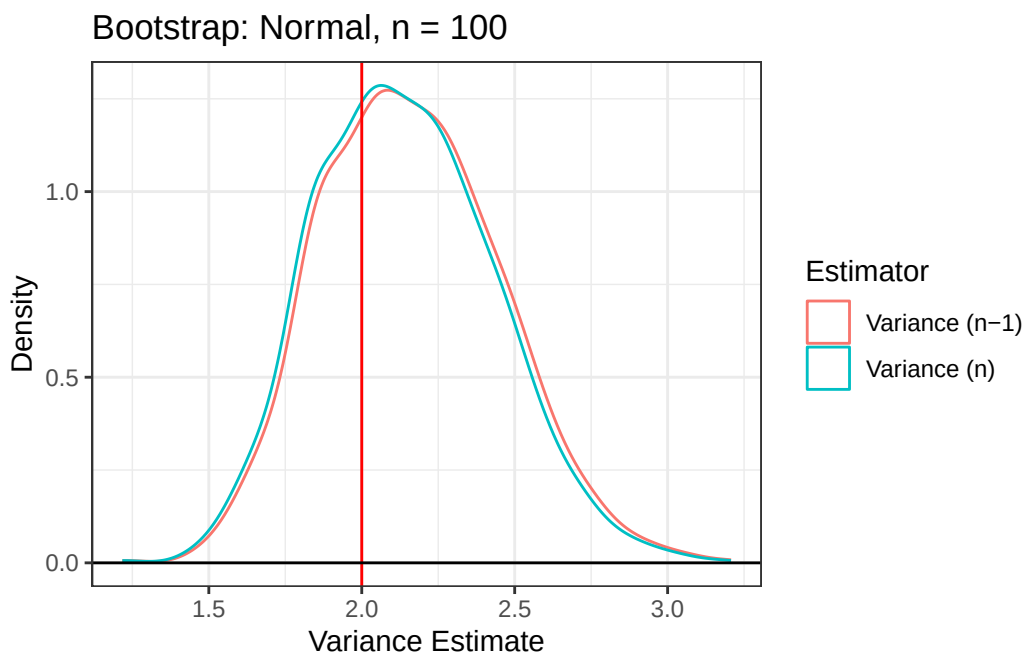
```
pois.boot$plot
```



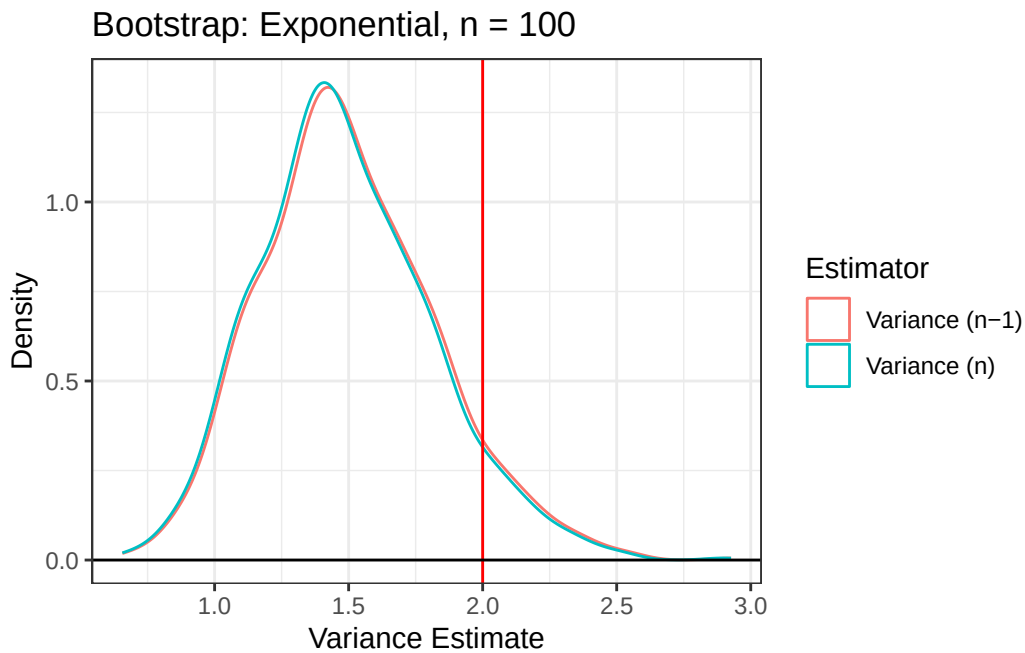

```
binom.boot$plot
```




```
norm.boot$plot
```




```
exp.boot$plot
```



Commentary: With a moderate sample size of $n = 100$, we notice that with bootstrap sampling, the distributions for our estimators exhibit more variance and are less predictable. This is especially noticeable when changing seeds. For example, for one seed (234567), we notice the estimators for the Poisson distribution are significantly more biased than the others, whereas the Exponential distribution exhibits this behavior when the seed is (2567). This demonstrates how bootstrapping is highly dependent on the particular random sample (which is what the seed establishes). Despite the increased variance and bias, the distributions still are "somewhat" close to the true value of 2 (though that description is somewhat liberal).

3 Executive Summary

Summarize the work you've done here to provide a brief (200-300 word) summary of the simulation and the results using your work in Lab 11. Your summary should be professional, clear, and all claims should be corroborated with statistical support.

Response: In the above lab, we compare two variance point estimators, S^2_{n-1} and S^2_n to identify why using the $n-1$ version is standard practice. We compare the two point estimators through three primary measures (listed in order of priority): (1) Bias, (2) Precision, and (3) Mean Squared Error (MSE). That is to say, we first would like to compare each estimator's bias: if both are unbiased, we compare the precision, if both are biased, we compare the MSE.

To obtain these values, we simulated 1,000 samples for four distributions: Poisson, Binomial, Normal, and Exponential. Each of these distributions is parameterized such that the true variance is 2, allowing for clear and consistent analysis. For each distribution, we also compared numerical and graphical summaries for multiple n values (20, 50, 100, 500).

The primary findings from this approach is that, regardless of distribution, S^2_n is consistently biased downward and S^2_{n-1} is consistently unbiased. This difference is more pronounced for small n: as n increases for each of these distributions, the biases of the two estimators converge toward 0. Note that because only S^2_{n-1} is unbiased, we do not need to analyze MSE. Furthermore, precision increases for both estimators with n, and the distributions for both estimators improve from right-skewed for small n to symmetric/approximately normal for large n.

After obtaining our primary results, we study if outcomes change under a bootstrapping procedure, and if so, how. The predominant result from this analysis was that bootstrap sampling produces significantly more variable estimator distributions that change wildly with the sampling seed. The bootstrap results still loosely depicted the behavior we exhibited before, but with less reliability.

Overall, this exercise confirms the advantage of using S^2_{n-1} over S^2_n, particularly when obtaining large samples is not

feasible.